

The impact of stellar wind and radiation on neutral knots around Wolf-Rayet star

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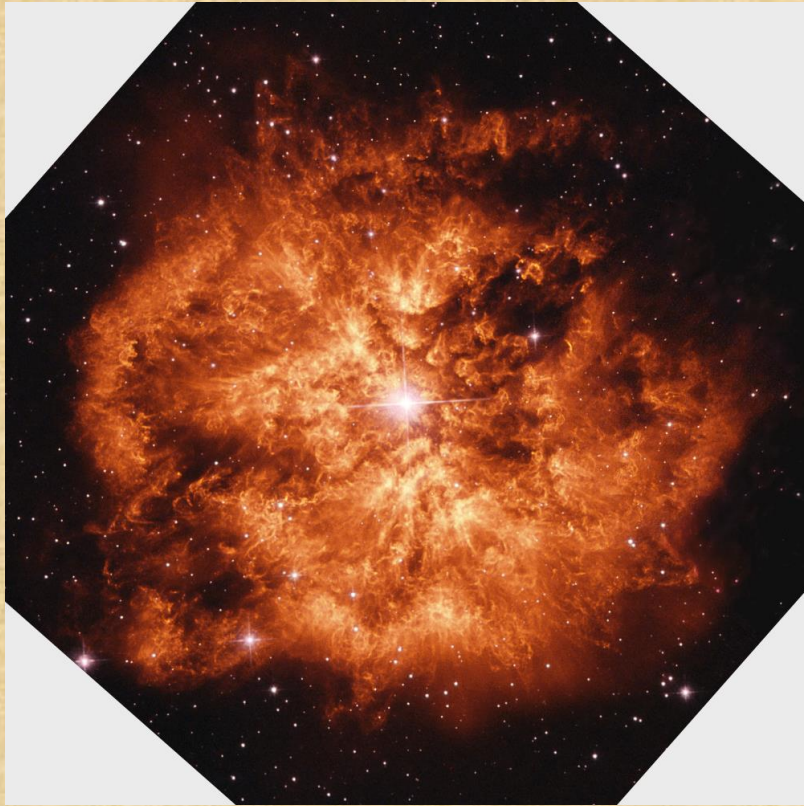
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Abstract

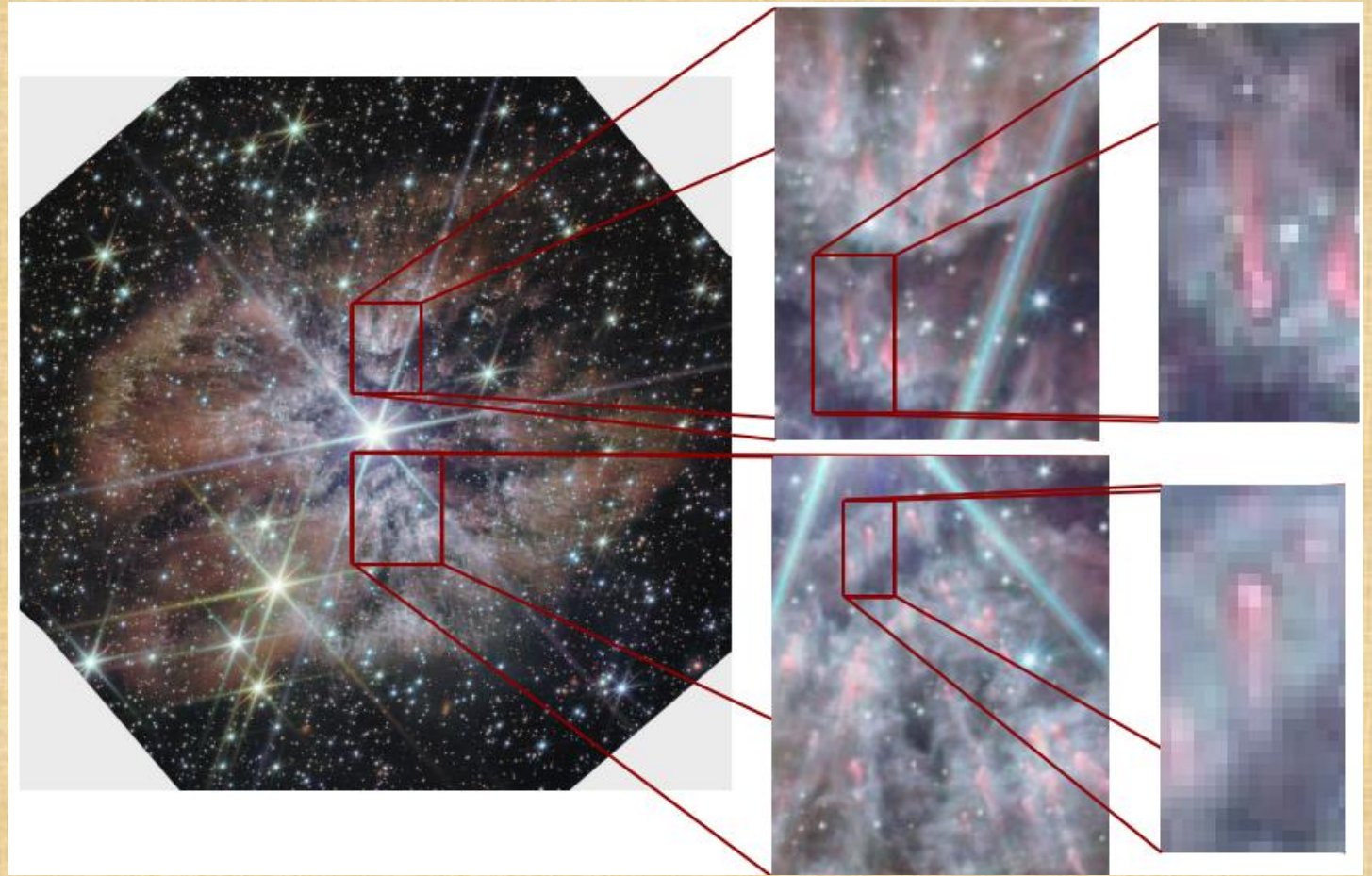
The circumstellar nebula M1-67 around the runaway Wolf-Rayet star Wr-124 contains hundreds of small neutral knots, as revealed recent JWST images. The ionized emission from the nebula shows an intricate pattern of shells and filaments, many of which appear associated with the knots but displaced toward the central star. We propose a simple model for the nebula in which photoevaporation flows from the irradiated surfaces of the knots interact with the stellar wind from the Wolf-Rayet star to form hemispherical emission shells. We test this model against JWST images and H alpha HST images of the nebula, confirming that the shells are in pressure balance with both the external stellar wind and the internal photoevaporation flow. We derived the three-dimensional distribution of the knots with the nebula and compare with observation of the knots kinematics, finding a close agreement. The model provides a physical explanation for the observed morphology of the nebula and knots, and suggest that the knots are hydrodynamical shielded from the stellar wind by photoevaporations flows.

Circumstellar nebula M1-67

Recent images JWST images reveal the presences of numerous neutral knots around WR-124, many of which associated with bright arcs of ionized gas emission located between the knots and the star. We measure the H alpha emission properties of these arcs from archival HST images.



HST WFPC2 H Alpha image of M1-67 nebula
(Grosdidier et al 1998)
Image credit: ESA/NASA/Judy Schmidt



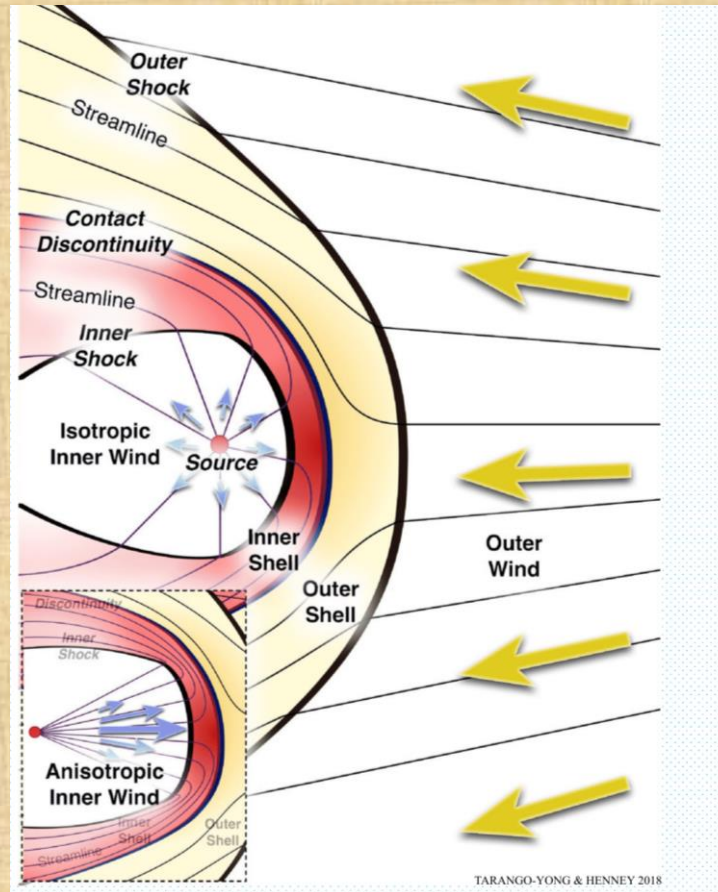
JWST NIRCAM image of M1-67, F444W, F335M,
F210M, F150W, F090W. Image processing by
Judy Schmidh
(<http://www.flickr.com/photos/geckzilla>)

Zoom on neutral knots (pink) and
interaction arcs (gray-white)

Steady State Hydrodynamic model

We propose that these emission arcs shock shell resulting from a supersonic wind-wind interaction between stellar wind from the WR star, and a transonic photoevaporation flow from the surface of the neutral knots that is induced by ultraviolet radiation from the star .

We employ a steady-state hydrodynamic model to calculate the equilibrium point where the RAM pressure of the two winds is in balance. This is also equal to the thermal pressure of the shocked shell, which can be observationally estimate from the HST images.



Catoon of the interaction between two supersonic flows. The inner wind is slow and dense, while the outer wind is fast end low density. Only the inner shell is expected to be visible. (Tarango-Yong & Henney 2018)

Steady State Hydrodynamic Model

The left image is the scheme of the two winds interaction. The knot is the orange circle, the thin red lines are the transonic photoevaporation flow. We considered that in the surface of the neutral knot the photoevaporation flow has a Mach number of 1, this part is dominated by thermal and hydrodynamic pressure

$$P_{tot} = P_{the} + P_{hyd} = n\bar{m}c_s^2 + n\bar{m}u^2 = \rho c_s^2(1 + M^2)$$

This photoevaporation flow is radial and travel until to have an equilibrium with the pressure of the shock shell, this pressure is only thermal pressure. In this part, the pressure of the photoevaporation flow fall a fraction that it was initially, so the pressure changes as

$$\frac{P}{P_{tot}} = \frac{\rho}{\rho_0} \frac{1 + M^2}{2}$$

If we consider the equation of conservation of mass

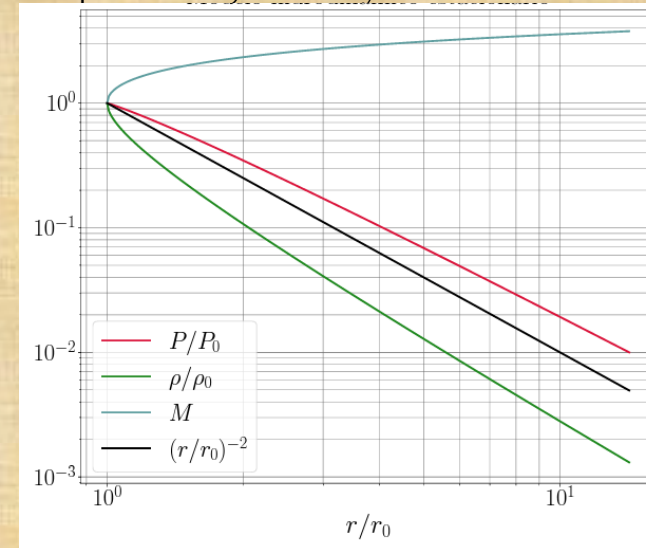
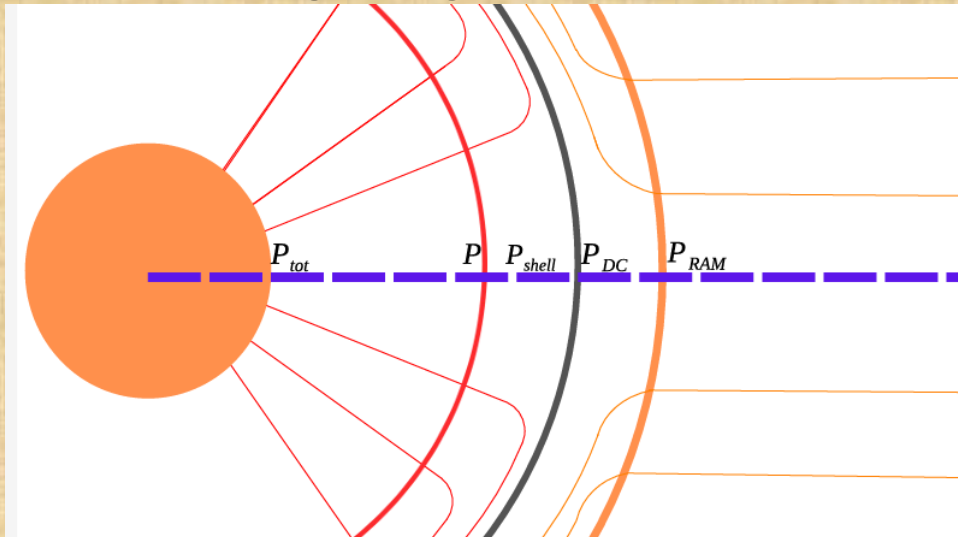
$$\rho r^2 M = \rho_0 r_0^2$$

And the isothermal Bernoulli equation

$$\frac{r}{r_0} = M^{-1/2} \exp\left(\frac{M^2 - 1}{4}\right)$$

We have our Model, the right image, in which we can see how changes the pressure, density and Mach number as function of r.

Scheme of the two winds interaction, thin red lines are the photoevaporation flow and the thin yellow lines are the stellar wind



Functions of pressure, density and Mach number normalized as function of r/r_0 .

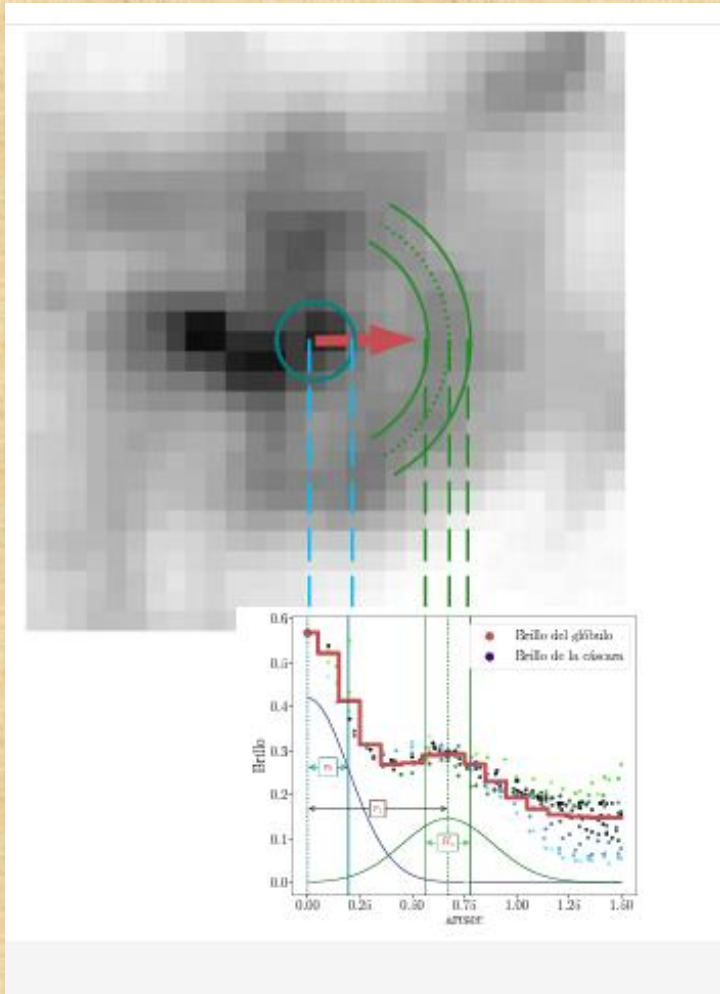
Observations

Once we distinguished the knots in the HST and JWST images, we fit two gaussians and a constant to the brightness profile to have the radius of the knot, the radius of the shock shell, the height of the shell, the brightness of the knot and shell. With this, we can obtain the density of the shell, which is about 1800 cm^{-3} , and compare the pressure of the shock shell

$$P_{shell} = \rho_2 c_s^2$$

With the RAM pressure of the stellar wind

$$P_{RAM} = \frac{\dot{M} v_\infty}{4\pi R^2}$$



Example of fit of two Gaussians and a constant to brightness profile. With this fit we know the size of the knot, the radius and height of the shell, and the brightness of the knot and the shell.

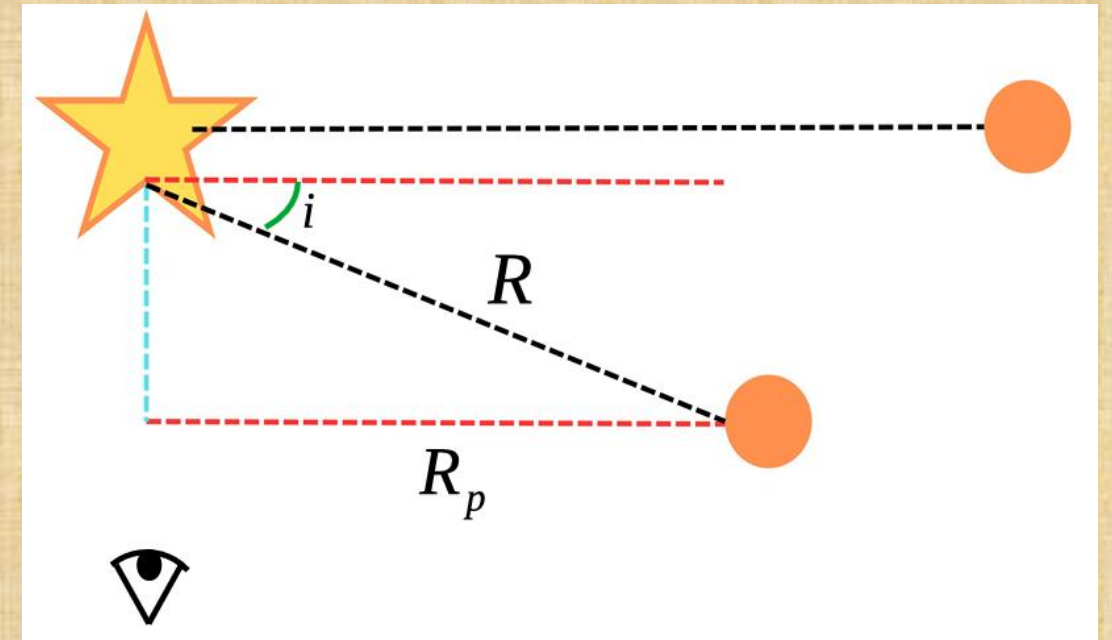
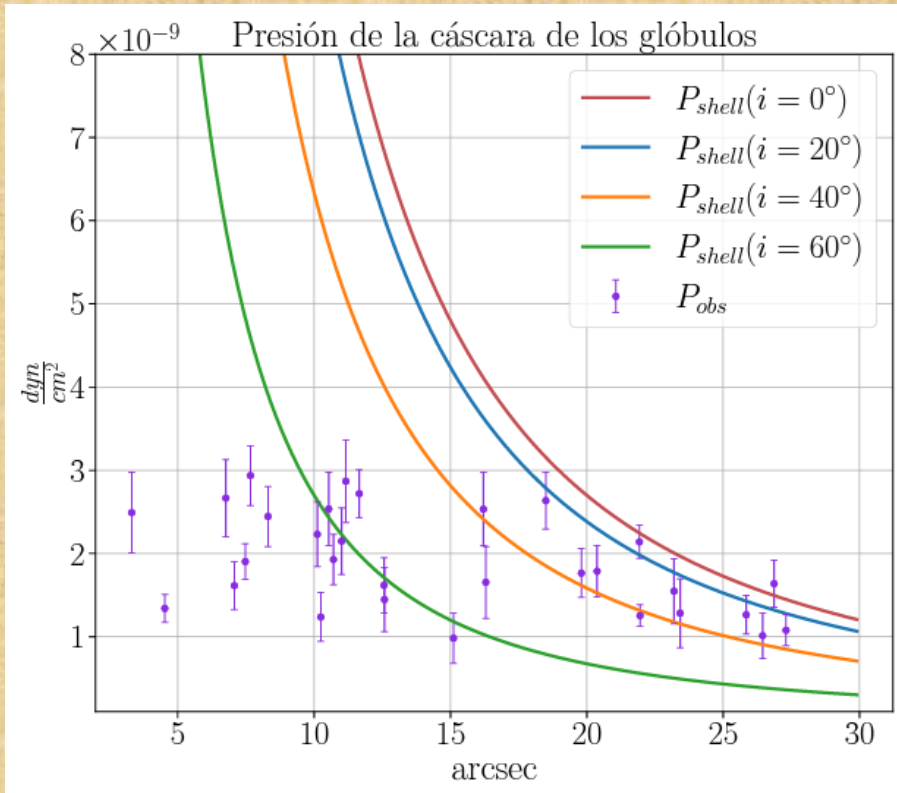
Comparison between the model and the observations

On left images we can see that the pressure of the shell are minor that the RAM pressure (red line), so we must correct by an inclination angel, as we can see on right image. With this consideration we have the real separation between the star and the knots, and also we now the angle inclination, so we have the three-dimensional distribution of the knots around WR star.

The correction by inclination angle are

$$R_{projected} = R \cos i$$

$$P_{shell}(i) = \frac{\dot{M} v_{\infty}}{4\pi R^2} \cos^{5/2} i$$



The pressure of the shell (purple circle) are equal to the RAM pressure (colors lines) if we considered an inclination angle.

Correction by inclination angle i

Comparison between the model and the observations

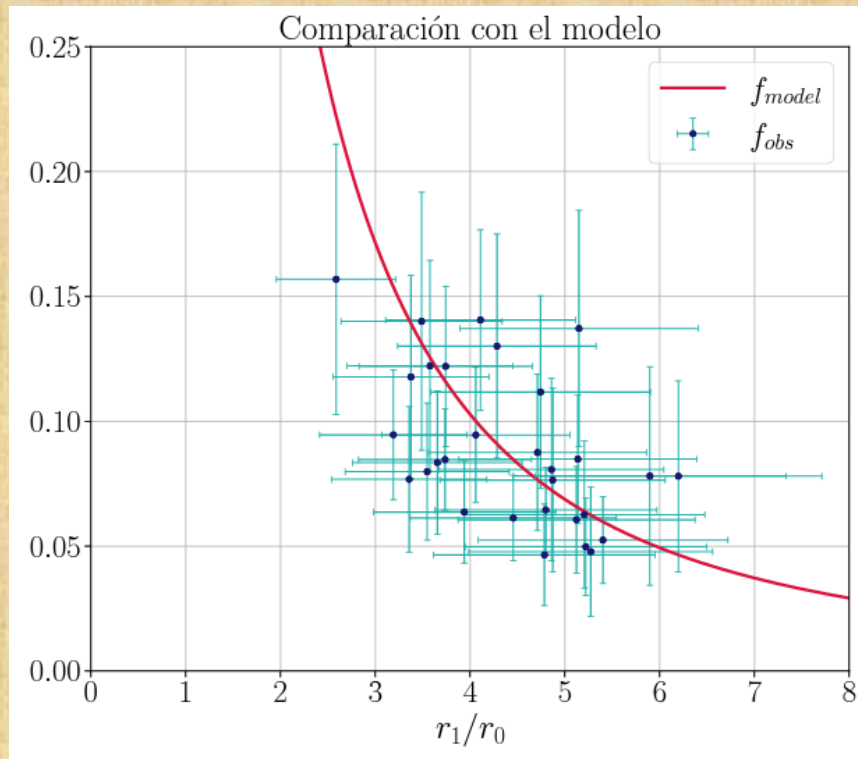
Also, we compare with the model predicted for the ratio between the pressures. We compare the pressure in the surface of the knots and the pressure of the shells as

$$\frac{1}{2} \sqrt{\frac{B_s/B_0}{l_s/l_0}}$$

Where B_0, B_s are the brightness in the knot and l in the shell, respectively, and is defined as

$$l = 2\sqrt{2rh}$$

The ratio of the pressures that we have of the observations agree with we predicted in our model as we can see in the image

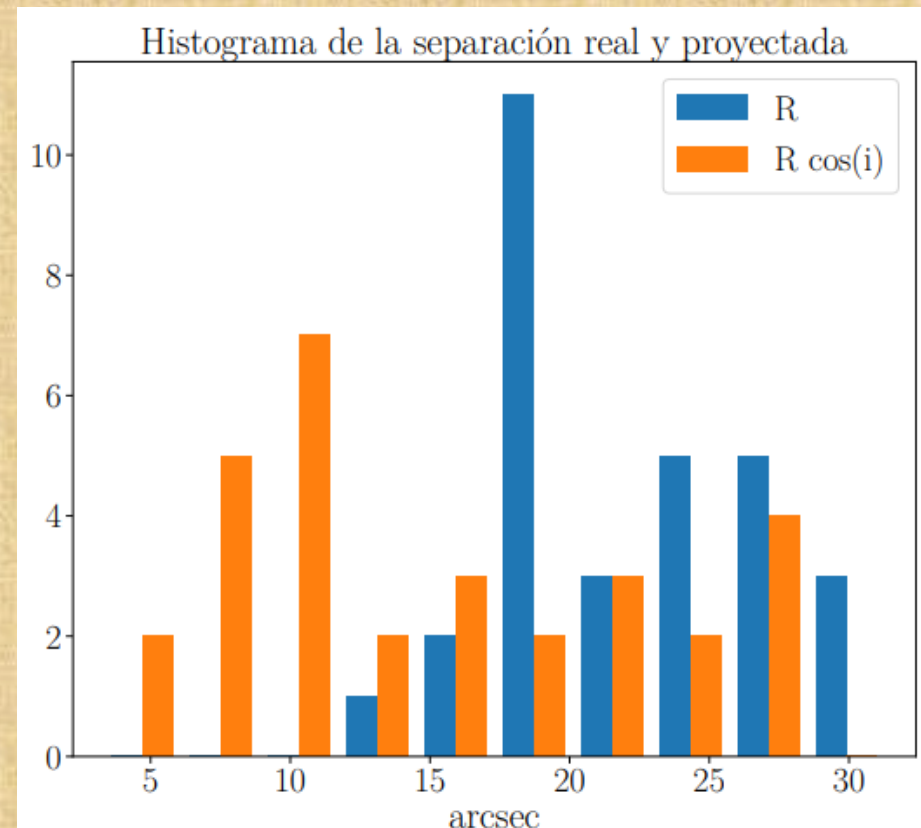


f_{obs} is the ratio between the photoevaporation flow pressure in the surface of the knots and the pressure of the shells obtained of the observation. f_{model} are the value that we predict with our model

Conclusions

Due the observations are agree with our model, this can explain the interaction between two supersonic flows. And we need to considerer that this model has no free parameters.

Also, we have three-dimensional distribution of this knots around the star, and this is according with another results (Zavala, et.al 2022)



Histogram od projected (orange) and real (blue) separation between the star and the knots.

Thank you for reading and follow the monarchs ☺